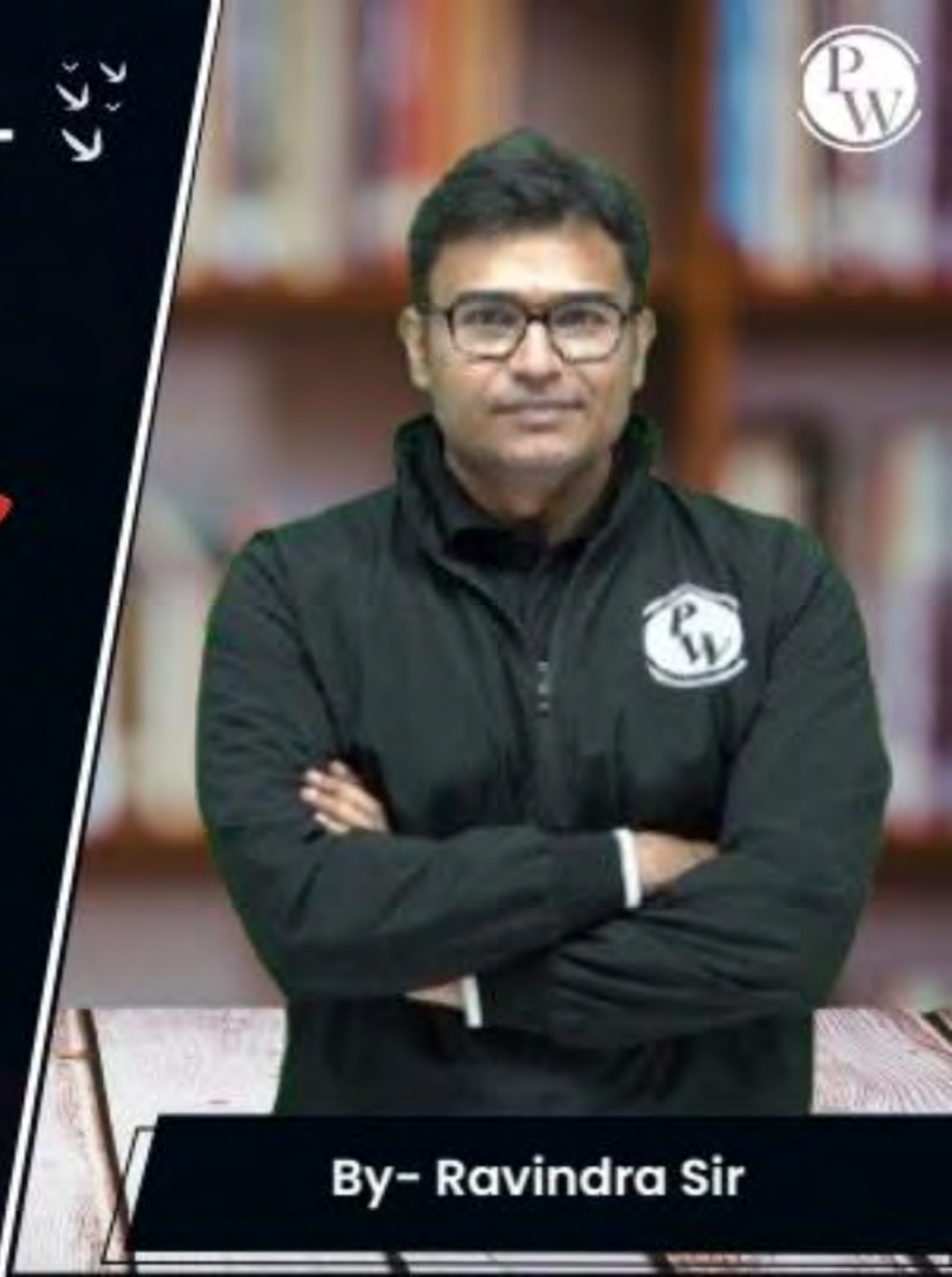
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 13



By- Ravindra Sir

Recap of Previous Lecture



- Topic — NFA → DFA
- Topic
- Topic

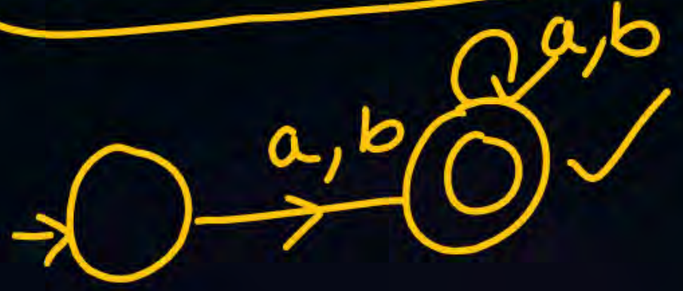
Topics to be Covered

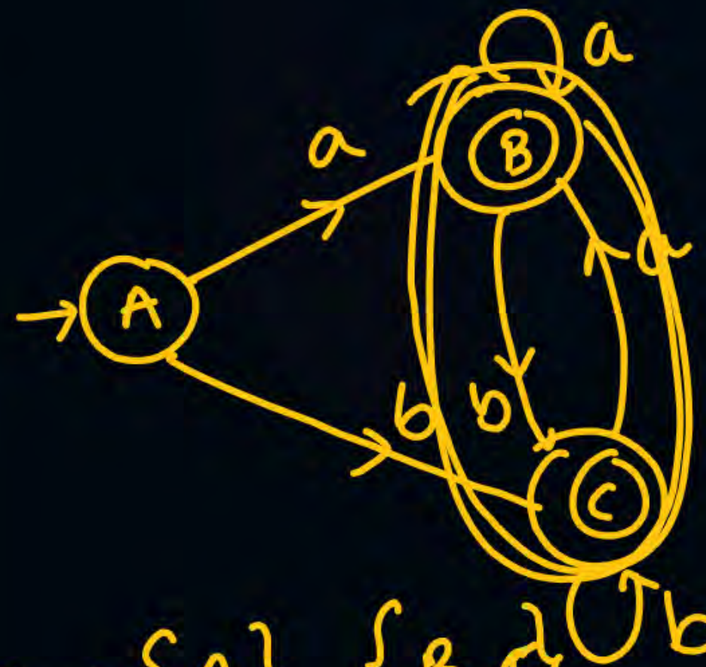


- Topic — DFA → Min DFA
- Topic — ∈ NFA → NFA
- Topic — Regular or not

$(a+b)^*(a+b)$

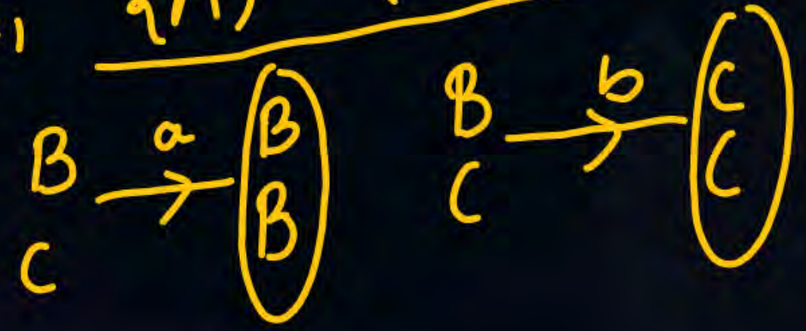
$(a+b)^*a + (a+b)^*b$





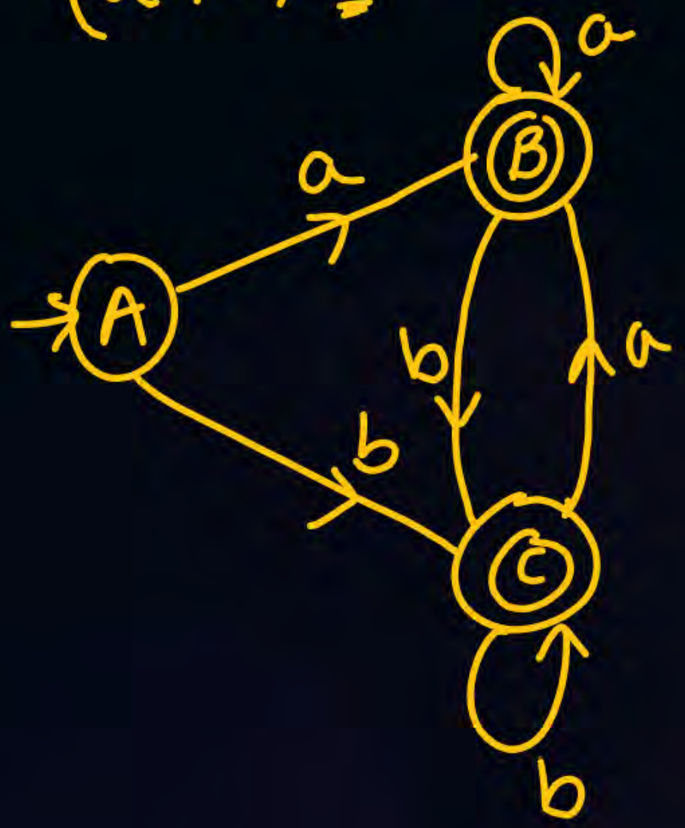
$\pi_0 \rightarrow \{A\}\ \{B,C\}$

$\pi_1 \quad \{A\}\ \{B,C\}$

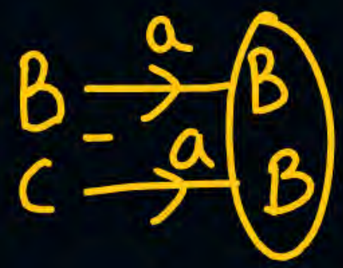


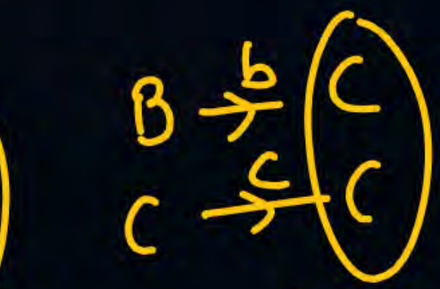
$(a+b)^*(a+b) = (a+b)^+ = (a+b)(a+b)^*$

$(a+b)^*a + (a+b)^*b$

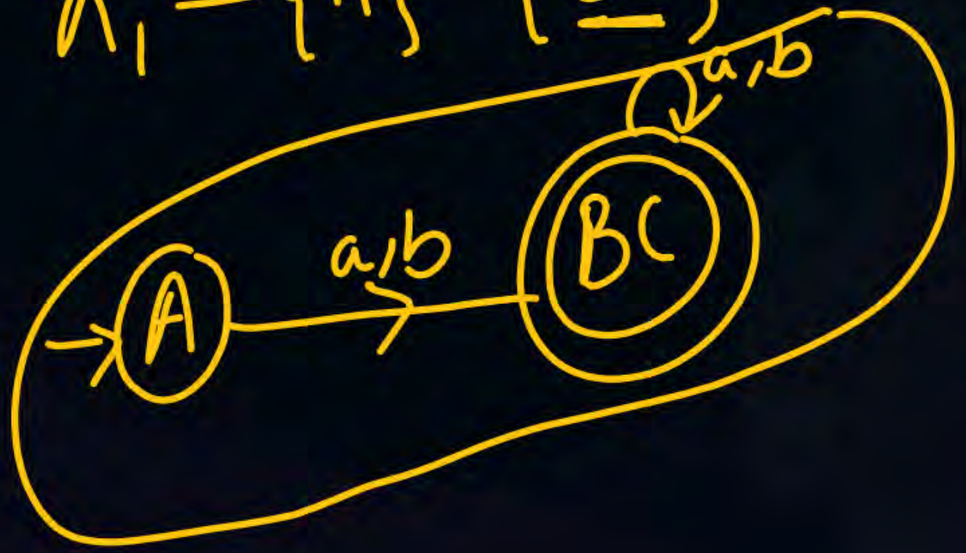


$\pi_0 - \{A\}\ \{B\ C\}$

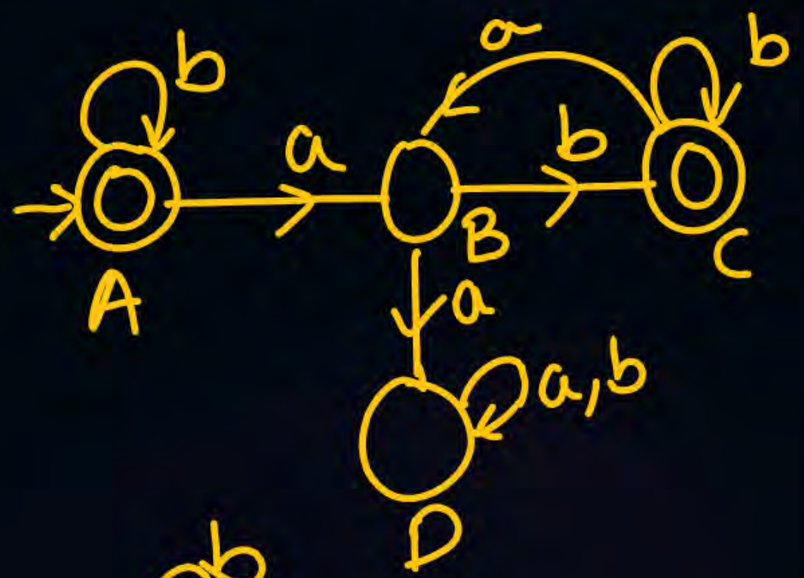


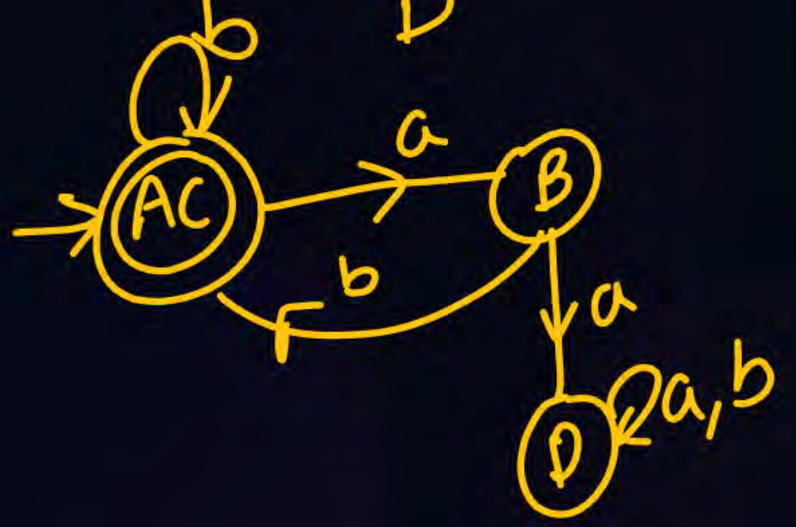


$\pi_1 - \{A\}\ \{BC\}$



a is always followed by ab



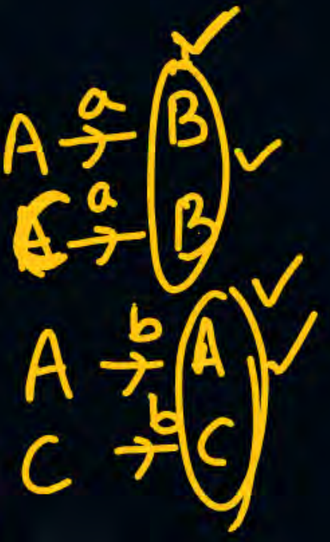


NF ✓ FS ✓

Π_0 - {B, D} {A, C}

B →a D ✓ B →b C ✓

D →a D ✓ D →b D ✓



Π_1 - {B} {D} {AC}

Π_2 - {B} {D} {AC}

0, 1, 2, 3, 4



π_0 {A D E} {B C}

A → 0 → B
D → 0 → E

A → 0 → B
E → 0 → E

D → 0 → E
E → 0 → E

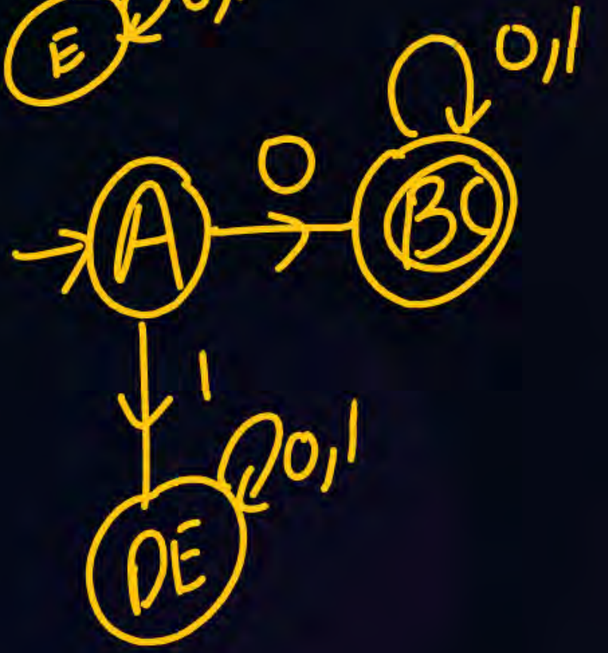
D → 1 → D
E → 1 → E

π_1 {A} {D, E} {B C}

B → 0 → C
C → 0 → C

B → 1 → B
C → 1 → B

π_2 {A} {D, E} {B C}



→ (Two states are equivalent if the both go to either final state & non final state on all strings.

→ (A, B are equivalent iff $\forall w \in \Sigma^*$ $\delta^*(A, w)$ & $\delta^*(B, w)$ both go to either final state & non final state

→ A, B are k equivalent if $\forall (w) \in \Sigma^*$ and $|w| \leq (k)$

$\delta^*(A, w)$ & $\delta^*(B, w)$ both go to either final & non final state

∈-NFA to NFA : (No of states will not change)



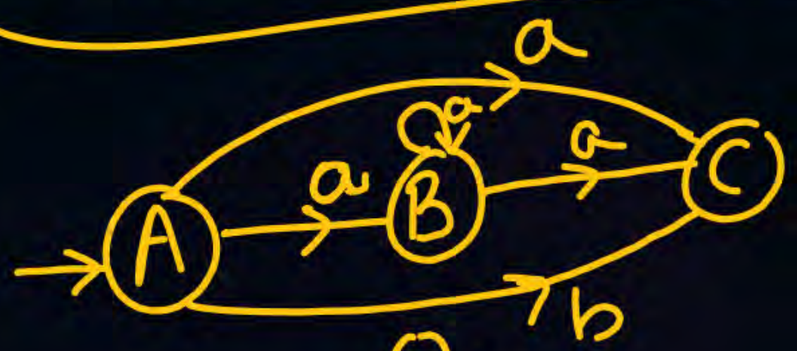
epsilon closure

$\delta^*(A, \epsilon) = \{A, B, C\}$

$\delta^*(B, \epsilon) = \{B, C\}$

$\delta^*(C, \epsilon) = \{C\}$

$A \xrightarrow{b} ?$



$A \xrightarrow{a} ?$

$B \xrightarrow{a} ?$

$\epsilon^* a \epsilon^*$

$B \to B \to B \to \begin{matrix} B \\ C \end{matrix}$

$B \to C \to \phi$

Identifying Regular languages:-

1) If a language is finite, it is always regular

$a^n b^n c^n d^n \mid n < 10^{10}$

↓

finite

$ww^R \mid |w| = 10$

finite

regular

2) Infiniti

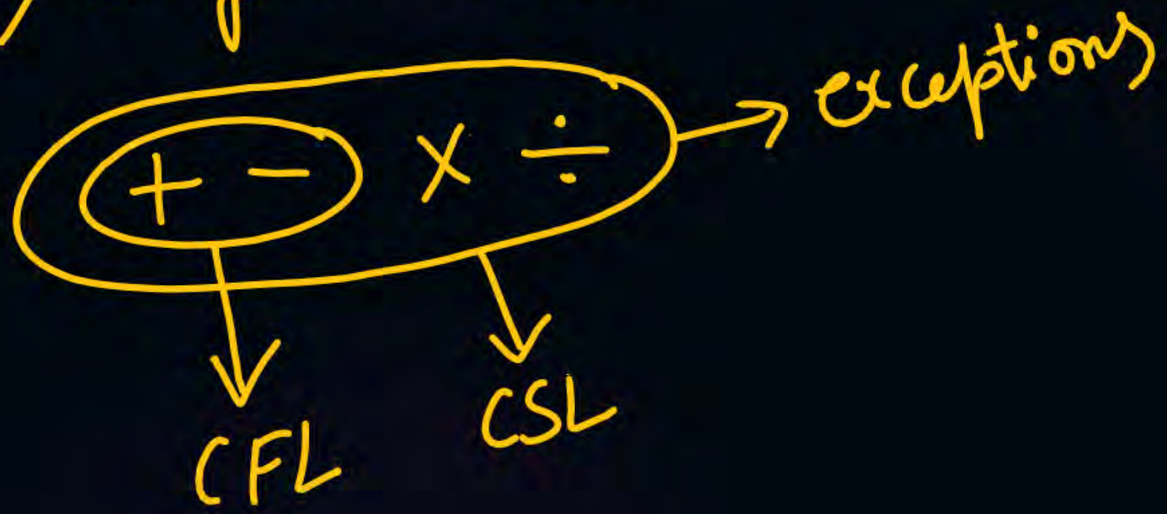
a) b) c) d)

2) a) non linear power → non regular (infinite language)

Non Reg:
- $a^{n^2}/n \geq 0$
- $a^{n!}/n \geq 0$
- $a^{3n^2}/n \geq 0$
- $a^{2n}b^{n^2}/n \geq 0$
- a^n/n is a perfect square

linear → Reg:
- a^n
- a^{3n+2}
- a^{kn+l}
- $a^{m+1}b^{n+2}$
- $a^{km+2}b^{ln+1}$

2) b) Complex rule



2) b)

$a^m b^n c^p / m+n=p$ → CFL / CSL ✓ ×



$a^m b^n c^p / n*m=p$ $n,m,p \geq 0$ → CSL XRL XCFL

$a^m b^n c^p / m+n+p=3$ → finite

0	0	3	
0	1	2	} finite → Reg
1	2	1	

$a^m b^n c^k / m/n = p$ → CSL ✗CFL ✗Reg

$m, n, p \geq 1$

$a^m b^n / \underline{m+n = p}$ ✓

m, n, p → variable

$m, n, p \geq 0$

CSL ✓
CFL ✓
✗Reg

$a^m b^n / m+n = 10$ ✓
$\neq 10$ ✓
≤ 10 ✓
< 10 ✓
> 10
≥ 10
} Reg

$L = \{a^m b^n / m+n \neq 10\}$ ✓

$\bar{L} = \{a^m b^n / m+n = 10 \cup (a+b)^* ba (a+b)^*\}$ ✓

$= Reg$

$L \quad \bar{L}$

$\bar{L} \quad L$

$a^m b^n$ / $2m+3n=10$ → finite → Reg

$a^m b^n$ / $5m+6n=100$ → finite → Reg

$a^m b^n$ / $m-n=10$
< 10
≤ 10
> 10
≥ 10
$\neq 10$
→ non Reg

$m-n=10$
$m = n+10$
Comparison

✓ CFL
✓ CSL

$a^m b^n / \underbrace{m \times n = p}_{\text{variable}} / n, m, p \geq 0 \rightarrow$ ✗ Reg
✗ CFL
✓ CSL

$a^m b^n / m \times n = 10$ ✓
< 10
≤ 10
> 10
≥ 10
$\neq 10$
} Reg

$a^m b^n$ / $\frac{m}{n} = p$ → CSL ✗CFL

3 variables

Note: CFG

$a^m b^n$ / $\frac{m}{n} = 10 \Rightarrow m = 10n$ → CFL

$a^m b^n$ / $\frac{m}{n}$ $= 10$, < 10, ≤ 10, > 10, ≥ 10, $\neq 10$ } Non Reg ✓CFL

2) c) Comparison

RL – 0 Comp

CFL – 1 Comp at a time

CSL – any no of Comp

$a^m b^n / m, n \geq 0$ $a^* b^*$ – Reg – 0 Comp

$a^{m+2} b^{2m+4} / m \geq 0$ – 1 Comp – CFL, CSL, XReg

$a^{n^2} b^{3n+2} / n \geq 0 \rightarrow$ CSL XCFL XReg

(+ -) (* /)

$a^n b^n / n \geq 0 \rightarrow$ CFL CSL

Same
- $a^n b^n c^n / n \geq 0$
- $a^m b^n c^p / m = n = p$
- $a^m b^n c^p / m = n$ and $n = p$

$\rightarrow$ 2 comp $\rightarrow$ CSL

Reg ✗

CFL ✗

$a^m b^n c^p$ / $m=n \wedge n=p$ (× Reg)

PDA

one Comp at a time → CFL → CSL

CFL

$a^m b^m c^n d^n / m, n \geq 0$

one Comp at a time — CFL — CSL

× Reg

2) d) String matching

ww^R / $w \in \Sigma^*$

↓

CFL → grammar

×Reg

CSL

$ww^R / w \in \Sigma^* \; |w| \leq 10 \rightarrow$ finite Reg

$ww / |w| \leq 10 \rightarrow$ finite $\rightarrow$ Reg

$ww^R / |w| \leq 10$ → finite Reg (0, 1, 2, ... 10)

≤ 10

< 10

$= 10$

≥ 10 infinite

> 10 non regular

$\neq 10$ CFL

$ww^Rw / w \in \Sigma^*$ — Regx
CFLX
CSL✓

CFL ✓

$w_1 w_1^R w_2 w_2^R \rightarrow$ CFL, CSL

one matching at a time

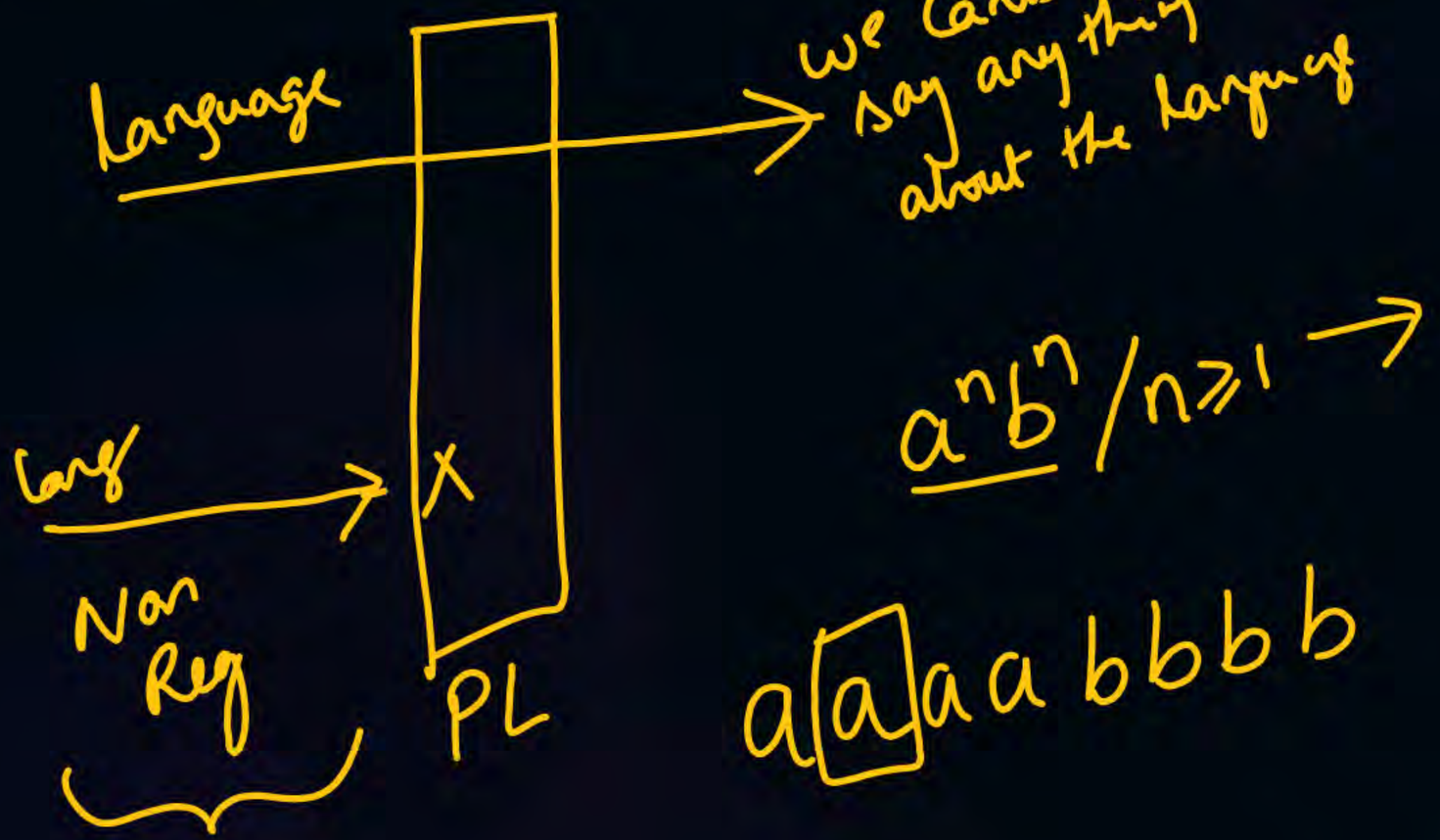
XReg

$w_1 \underbrace{w_2}_{w_1^R} \underbrace{w_3}_{w_1^R} w_4 \;/\; w_1, w_2, w_3 w_4 \in \Sigma^*$

$w_2 = w_1^R$ & $w_3 = w_1^R$



Pumping lemma :- Negative test



$a^n b^n / n \geq 1 \rightarrow$

a a a a b b b b

X Pumping lemma ✓

$a\underline{a}\underline{a}\underline{a}\underline{a}\underline{b}\underline{b}\underline{b}\underline{b}b$ X

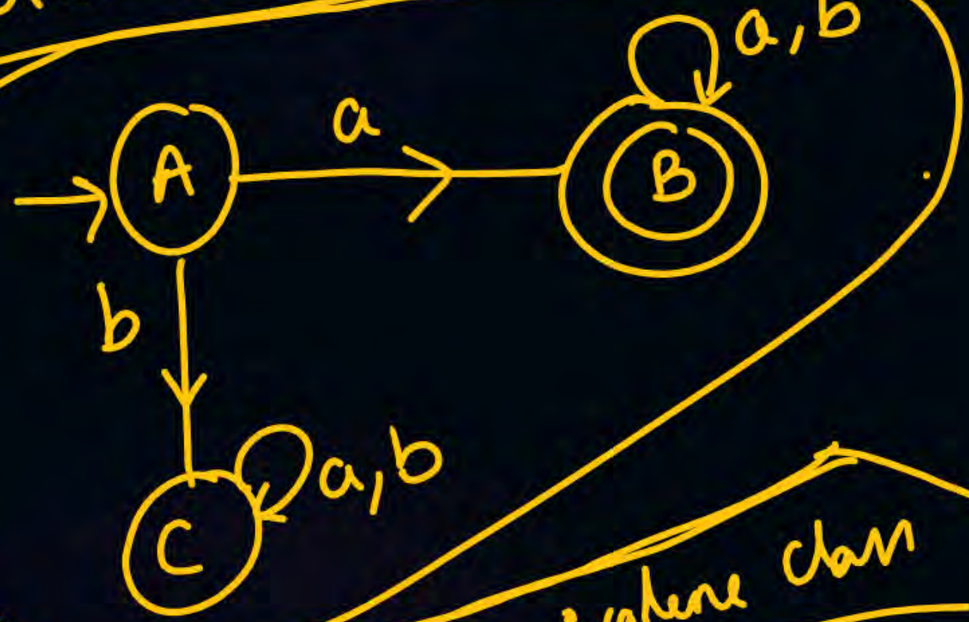
X Reg

Theorem:- Let L be an infite Reg language. Then there exists Some positive integer m such that $w \in L$ with $|w| \geq m$ Can be decomposed as $w = xyz$ with $|xy| \leq m$ and $|y| \geq 1$ Such that $w_i = xy^i z$ is also in L for all

$i = 0, 1, 2, \ldots$



Myhill-Nerode theorem :-



min DFA

$L_A \rightarrow \{\epsilon\}$ → equivalence class

$L_B \rightarrow \{a, aa, ab, aba \ldots\}$ – EC

$L_C \rightarrow \{b, ba, bb, bab \ldots\}$ – EC

$L_A \cup L_B \cup L_C = \Sigma^*$

$L_A \cap L_B = \phi$

$L_B \cap L_A = \phi$

$L_B \cap L_C = \phi$

myhill nerode theorem :-

It is based on minimal DFA

|equivalence classes| = |states in min DFA|

If equivalence classes are finite then we can build FA.



→ If a language is regular, the no of myhill nerode equivalence classes is finite.

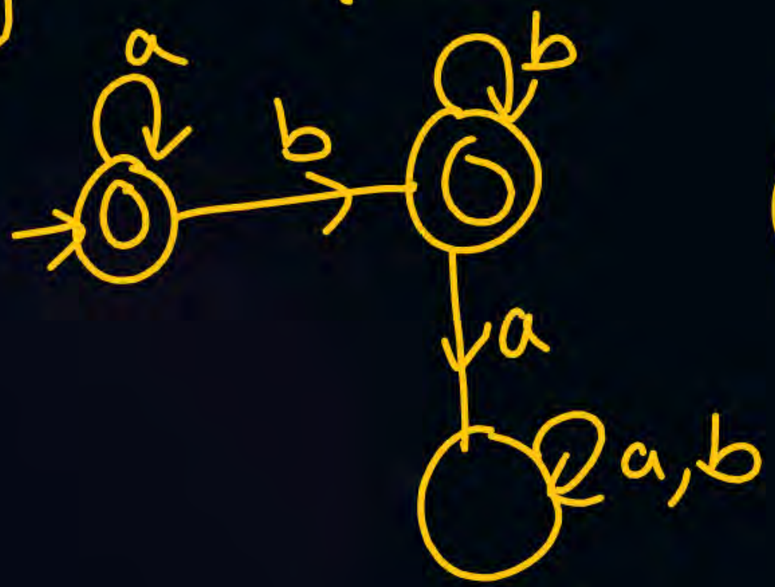
Reg → min DFA → States finite

↓

EC finite

→ Two strings x, y are equivalent if $\delta^*(q_0, x) = \delta^*(q_0, y)$

→ How many MN equivalence classes are in a^*b^*



3 ✓

min DFA

$a^n b^n / n \geq 0$, how many MN equivalence classes

non Regular

FAX

∞ equ class

X

Finite Auto

↓

finite

→ If $E_1, E_2 \cdots E_n$ are n equi classes

1) $|E_i| \geq 1$ (atleast one string)

min DFA 

2) $i \neq j \Rightarrow E_i \cap E_j = \phi$ (A string will not be present in two equivalence classes)

3) $UE_i = \Sigma^*$


RE
Rec
CSL
CFL
DCFL
Reg
Σ^*

Closure properties

	Reg	DCFL	CFL	CSL	REC	RE
$\cup$	✓	✗	✓	✓	✓	✓
$\cap$	✓	✗	✗	✓	✓	✓
L^c	✓	✓	✗	✓	✓	✗
$*$	✓	✗	✓	✓	✓	✓
$\cdot$	✓	✗	✓	✓	✓	✓
L^R	✓	✗	✓	✓	✓	✓
$h(L)$	✓	✗	✓	✓	✓	✓
$h^{-1}(L)$	✓	✓	✓	✓	✓	✓

✓ - Sure
✗ → may / may not - ✗

$L = \{a^m b^n c^p / m = n \text{ or } m = p\}$

L

CFL

CSL

closed

THANK - YOU